Review Sheet 3: Managing Risks (The Newsvendor Model)[1]

1. Matt Herron*

Matt Herron is the chief buyer at Investment Clothiers, a retail store known for excellence in apparel. It is time to order merchandise for the Christmas season. During a recent trip to Hong Kong, Matt spotted a particular men's overcoat that he expects will sell very well. Based on past experience, Matt expects demand to range from 100 to 400 overcoats, with probabilities that are as follows:

Estimate of Season's Demand	*Probability*
100	0.1
200	0.4
300	0.4
400	0.1

The total cost to Investment Clothiers would be $60 per overcoat, and the retail price is estimated at $110 per overcoat. Any overcoats left over after the Christmas season are expected to be sold at $40 each.

a) *Calculate the expected profits for each of the following four cases— when Matt buys 100, 200, 300 or 400 overcoats. How many overcoats should Matt buy if he wishes to maximise expected profits over the coming Christmas season?*

b) *Now solve the problem using marginal analysis.*

c) *The assumption of a discrete demand distribution is a bit artificial. Solve the problem using a normal distribution for demand (with the same mean and standard deviation as the discrete distribution of the original problem description). Does this change make much difference?*

[1] Problems are rated for difficulty: * = Relatively easy, for first review of concepts; ** = Challenging, assumes good understanding of the concepts; *** = Advanced, assumes full understanding of the concepts and creativity in application.

2. Eloïse's Pastry**

Eloïse is baking fresh pastry at several shops throughout the Leipzig area. Each pastry sells for 4.00 € and costs 1.80 € to make. Each one not sold at the end of the day is sold the next morning for 1.00 € as „yesterday's bargain pastry".

Daily demand is distributed as follows:

Demand	100	150	200	250	300	350	400
Probability	.05	.10	.20	.25	.25	.10	.05

a) *How many pastries should Eloïse bake each day? How many of those will she expect to sell the same day, how many the next day, and what is her expected daily profit?*

b) *Resolve part a), this time using a normal distribution for daily demand, with the same mean and variance as in part a)*

3. Selection of electricity generating capacity (with input from Eric Denardo)***

You are considering the decision on the quantity and type of electricity generating capacity to install for meeting the uncertain demand during the next quarter. Once chosen, this capacity remains fixed for the foreseeable future.

Demand over this period is uncertain with a mean weekly demand rate of 1,000,000 units (of electricity) per week and a standard deviation of weekly demand of 200,000 units. To play it safe you have decided to install sufficient capacity to be able to supply 1,600,000 units of electricity per week. This amount of capacity would indeed cover most reasonable demand scenarios (up to 3 standard deviations above the mean).

The question you are facing concerns the mix of plants that will make up your capacity. You have two types of plant: *base* load and *peak* load plants. So the sum of the capacities of your base load and peak load plants should be 1,600,000 units of electricity per week. Base plants are more expensive to install, but cheaper to operate. Peak load plants have the opposite characteristic: they are more expensive to operate, but require a smaller capital outlay to bring on line.

The weekly capital cost for a unit of base capacity is 3.3 €, that of a unit of peak load capacity is only 2.5 €. Capital costs are to be paid *regardless of usage*. Operating costs are to be paid only as a function of usage. The estimated operating cost for delivering a unit of electricity in a given week with base load is equal to 1.7 €, whereas the estimated cost using peak load is equal to 4.0 € per unit delivered.

a) *What is, given the above parameters, the optimal mix of peak and base load capacity? (Note that one additional unit of base load capacity saves you one unit of peak load capacity).*

b) *What is the expected weekly cost of electricity supply given this mix?*

4. Storewars**

Consider the situation by a manufacturer of play-stations, SUNNY, who is facing Christmas demand. SUNNY sells the play-stations through a large retail chain, TOYS-are-MINE, at a manufacturer (wholesale) price of 500 € per play-station. SUNNY's unit manufacturing cost is 350 € per play-station.

TOYS-are-MINE offers the play-stations for sale at its retail stores at the retail price of 800 € per station. Its demand forecast for the upcoming Christmas season is for 200 000 stations, with a standard deviation of 50 000. The retail chain has to order ahead of the season. Any play stations ordered will be delivered in time for the Christmas sales season, but no re-supply is possible during the sales season.

Any play-station that will not be sold during the upcoming Christmas will be sold after the season at a discount, for a price of 300€ per station.

a) *What is the optimal order quantity for TOYS-are-MINE, and what are its expected profits on this item?*

b) *What is the optimal order quantity on this item, if the retailer and the manufacturer were integrated into a single company?*

c) *Compare the expected profits generated by the integrated supply chain (which you computed in your answer to sub-question 2), with that of the chain formed by the two independent firms, SUNNY and TOYS-are-MINE (in the latter case, we have assumed in sub-question 1 that the order quantity was determined by the retailer). Which arrangement generates the highest expected profits for the supply chain? Why? What other major differences do you observe between these two inter-organisational arrangements?*